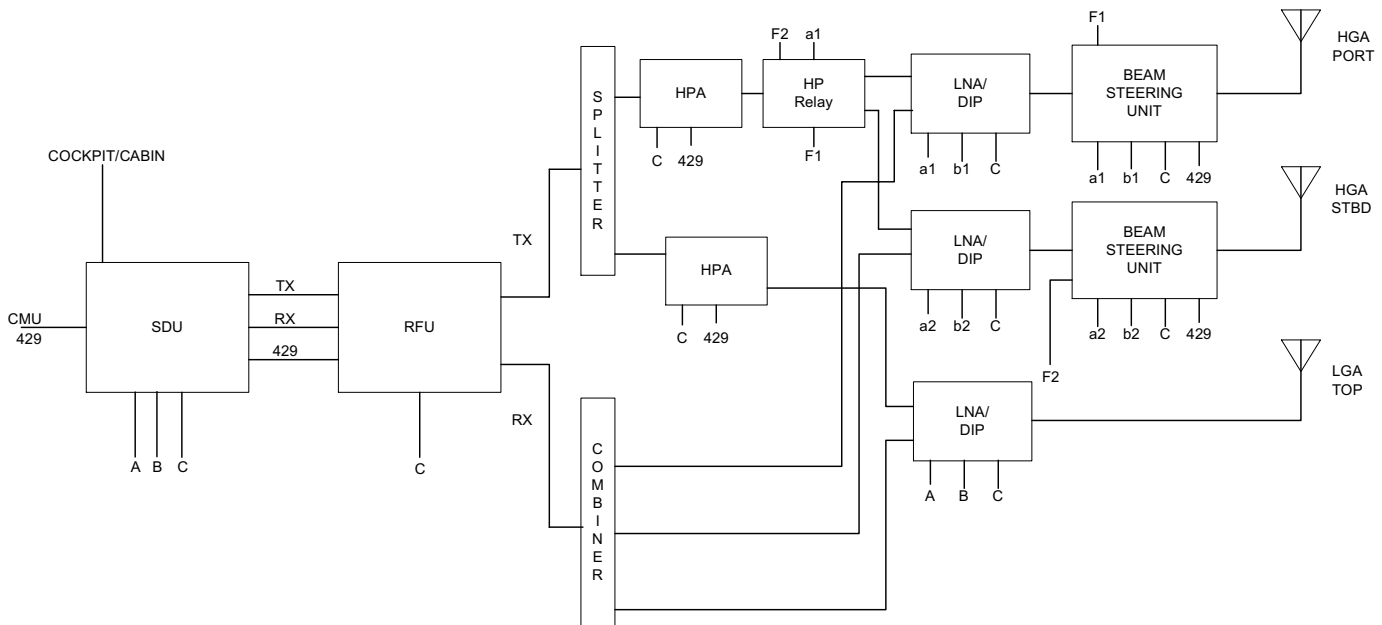


ATTACHMENT 1-2 ANTENNA CONFIGURATION



LEGEND

A/a1/a2 - LNA ON/OFF a1 is also HP Relay control

B/b1/b2 - LNA BITE

C - 115 Vac

CMU - COMMUNICATIONS MANAGEMENT UNIT

STBD - STARBOARD

429 - ARINC 429 DATA BUS

LNA/DIP - LOW NOISE AMPLIFIER/DIPLEXER

HGA - HIGH GAIN ANTENNA

HPA - HIGH POWER AMPLIFIER

LGA - LOW GAIN ANTENNA

RFU - RADIO FREQUENCY UNIT

SDU - SATELLITE DATA UNIT

F1/F2 HP Relay BITE

Notes:

1. This block diagram does not necessarily include every interface.
2. See the Antenna Subsystem Control Interface drawing of this configuration (located after the Avionics Block Diagrams within this attachment) for complete details of Beam Steering Unit Interfaces with the SDU, HPA, LNA/DIP, and HP Relay.
3. The low gain antenna and associated components are optional to this configuration.

Figure 1 – Side Mounted Phased Array Configuration with High Power Relay Option

FWD antenna is VHF-C and
AFT antenna is ELT

This image shows 9M-MRQ
with the single top-
centreline HGA radome, no
LGA and no side mount
HGA patches.

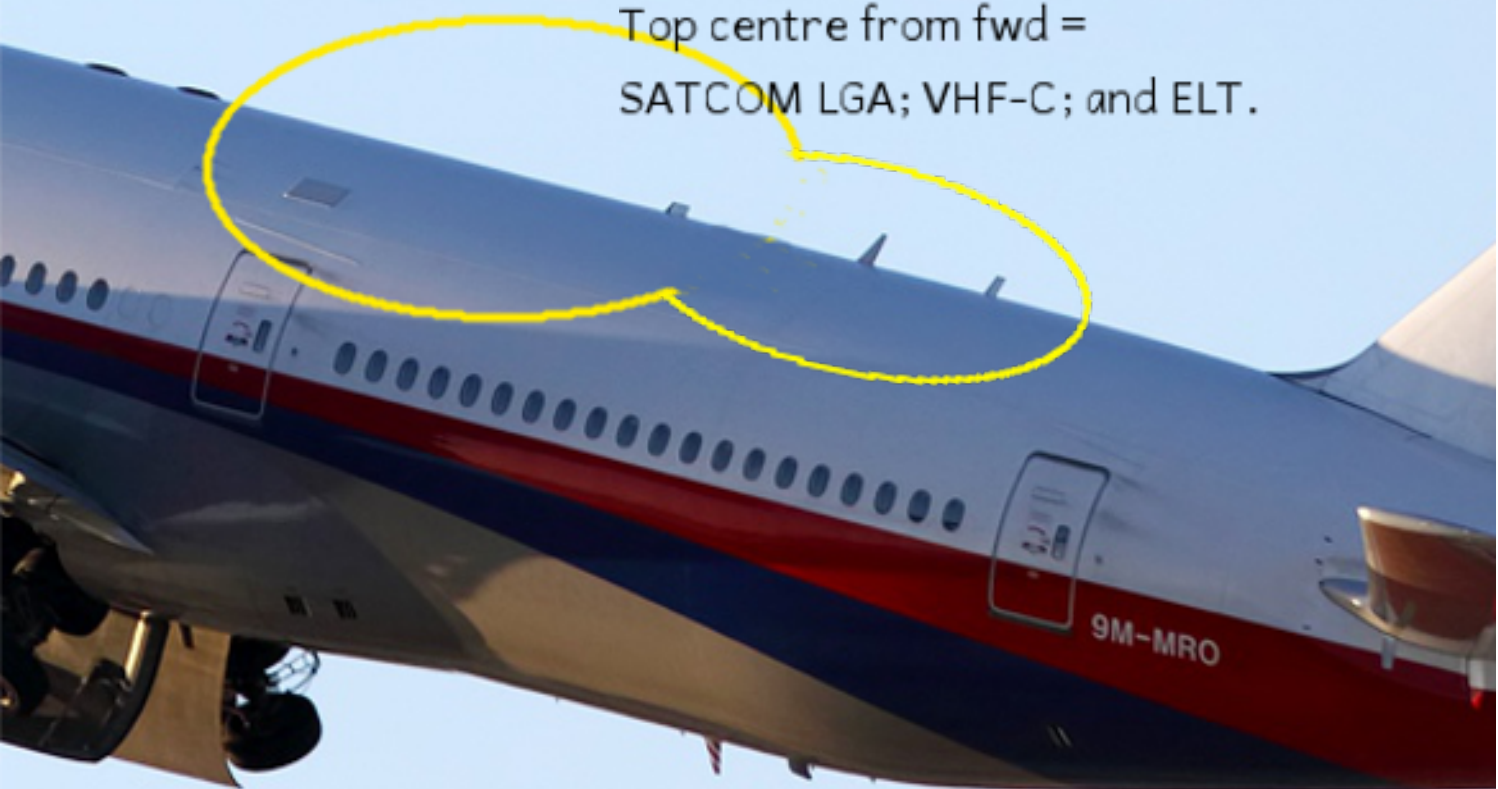


9M-MRO Antenna

Above door 3 = HGA side patches

Top centre from fwd =

SATCOM LGA; VHF-C; and ELT.





DESCRIPTION

S65-8282-101: Low-gain satcom antenna covering Inmarsat frequency range of 1530 to 1660 MHz, designed to meet SB200 Class 15 service. The broadband coverage of the antenna also includes frequencies of 1616 to 1626.5 MHz for Iridium Satellite Services voice and data applications. The unit provides over 90% sky coverage.

Small, lightweight, low-drag antenna is DC-grounded and hermetically sealed.

The antenna is approved for Inmarsat Aero-L and Aero-C (store-and-forward) applications. May be used as a back-up antenna for the high-gain ARINC 741 antenna.

The S65-8282-101 is also approved as Iridium Compatible Equipment (ICE).

NSN: 5985-01-457-5077

FEDERAL & MILITARY SPECS: ARINC 741, FAA PMA, DO-160D, MIL-HDBK-5400T, MIL-STD-810, MIL-E-5400

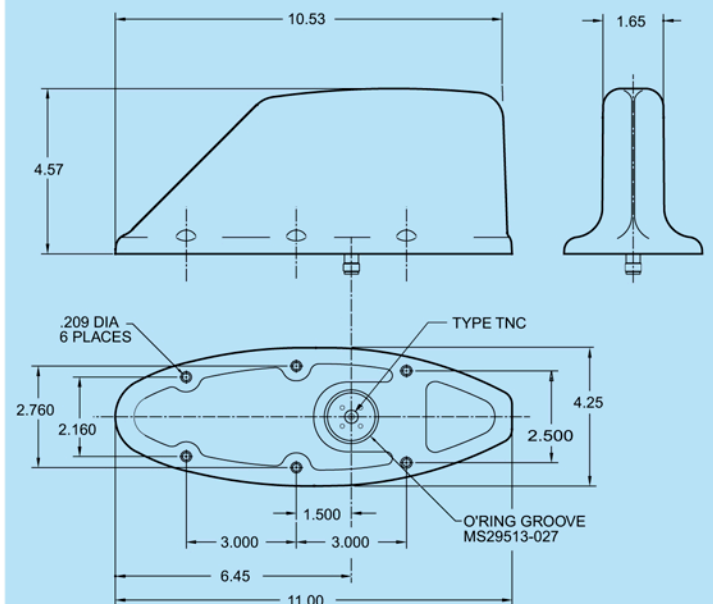
PERFORMANCE

G/T	≥ -21.5 dB/K
Minimum Sky Coverage: 95.4%	
A/R	≤ 3.6 dB
Maximum Sky Coverage: 93.6%	
Gain	≥ 2.7 dBic
Minimum Sky Coverage: 98%	
Reference	
Cable Loss: 0.2 dB	
LNA Noise Figure: 1.2 dB	



SPECIFICATIONS	FOR REFERENCE ONLY
MODEL	S65-8282-101
ELECTRICAL	
Frequency	1530-1559 MHz
	1626.5-1660 MHz
VSWR	1.5:1
Polarization	RHCP
Impedance	50 ohms
Power Handling	60 watts CW
Lightning Protection	DC grounded
MECHANICAL	
Weight	1.5 lbs. (0.68 kg Max.)
Height	4.57 in.
Length	11 in.
Width	4.25 in.
Material	6061-T6 aluminum / thermoset plastic
Finish	Skydrol-resistant enamel
Connector	TNC
Drag	1.1 lbs. Mach .85 @ 35,000 ft.
ENVIRONMENTAL	
Temperature	-65°F to +185°F (-55°C - +85°C)
Vibration	10 Gs
Altitude	70,000 ft.

OUTLINE DRAWING



S65-8282-101

ISO 9001:2000 and EN/JISQ/AS9100:2004 Registered



AIRLINK® High-Gain Antenna System (HGAS)

AIRLINK® Classic HGAS Systems can be upgraded to support the SwiftBroadBand (SBB) services.

Specifications Part Number 513738-5XX

Antenna Array	2 each
Frequency Range	1.525 to 1.559 GHz rcv., 1.6265 to 1.6605 GHz Xmit.
Size	16 in (407 mm) x 32 in (813 mm) x 0.375 in (9.5mm)
Weight	15.6 lb (7.1 kg)
Input Power	None required (Passive)

Specifications Part Number 513739-506-99

Beam Steering Unit (BSU)	2 each
Function	Provides the steering commands to the antenna
Size	3.5 in (89 mm) x 10.4 in (264 mm) x 13.5 in (343mm)
Weight	16.4 lb (7.4 kg)
Input Power	115 Vac, 400 Hz, <500 mA

Specifications Part Number 511510-500

Diplexer/Low Noise Amplifier (DIP/LNA)	2 each
Function	Separates signals into the transit and receive RF channels. Amplifies receive signals from the antenna.
Size	2.0 in (51 mm) x 7.8 in (198 mm) x 11.1 in (282 mm)
Weight	6.5 lb (3.0 kg)
Input Power	115 Vac, 400 Hz, <100 mA

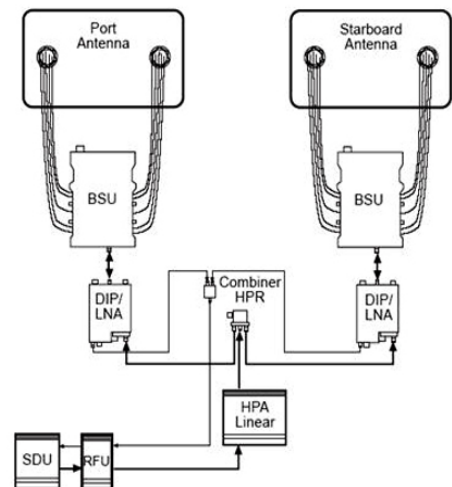
Specifications Part Number 301706-500

High Power Relay (HPR)	1 each
Function	Two-way RF switch with status contacts
Size	1.2 in (30.5 mm) x 1.9 in (48 mm) x 3.1 in (79 mm)
Weight	0.75 lb (0.34 kg)
Input Power	115 Vac, 400 Hz, 43.5 mA

Specifications Part Number 300114-500

Combiner	1 each
Function	Combines RF signals received from the 2 antennas
Size	2.0 in (51 mm) x 2.0 in (51 mm) x 0.8 in (20 mm)
Weight	0.75 lb (0.34 kg)
Input Power	None required (Passive)

AIRLINK® HGAS is a side-mounted, conformal electronically steered phased array. The HGAS is comprised of two antenna assemblies located on the aircraft exterior at nominally 45 degrees on either side of the aircraft. This configuration provides coverage of 360 degrees in azimuth and up to 210 degrees in elevation – 57 percent greater than a single top-mount antenna configuration. This ensures high reliability and provides superior coverage for all latitudes and aircraft maneuvers. The aerodynamically efficient design produces the lowest drag of any SATCOM antenna system. The configuration places all of the active electronics inside of the aircraft for high reliability.



**Ball Aerospace
& Technologies Corp.**

Agility to innovate. Strength to deliver.

THE AIRLINK® HIGH GAIN ANTENNA SYSTEM

Patrick Westfeldt, Jr.

John J. Konrad

Ball Aerospace and Communications Group

Broomfield, Colorado 80038

ABSTRACT

Developed first for commercial telecommunications, the AIRLINK® High Gain Antenna System has become the market leader in commercial aircraft installations. Two side-mounted phased arrays are employed on a single aircraft, using conformal passive apertures with active electronics modules mounted inside the skin. Coverage is achieved throughout the INMARSAT coverage region, and the system meets all ARINC, FAA, and INMARSAT requirements. The commercial system has also been reconfigured as a military secure telecommunications link (STU-III). The reconfigured system was demonstrated successfully in May of this year, resulting in the development of a mobile demonstration vehicle and a number of probable system deployments.

The passive conformal aperture is a multilayer printed circuit antenna and feed system less than 0.4 inches in overall thickness. Radiating elements are microstrip patch antennas covering both transmit and receive bands, while the feed is a buried microstrip structure sharing the radiating aperture. RF interfaces from the aperture elements to the electronics modules are accomplished with two custom cable bundles. The system architecture achieves balance between the competing demands of RF performance, reliability, modularity and low production cost to be competitive in commercial markets. Antenna pattern and system performance are presented in this paper.

1.0 INTRODUCTION

As an example of dual use technology, the AIRLINK® system may be unusual in that it was developed first at Ball Aerospace for commercial and general aviation. The development actually began in 1983 with an INMARSAT feasibility study, followed by system and hardware development funded internally and under a second INMARSAT contract. An important milestone was reached in 1991 with a business alliance with Collins Air Transport Division, whereby Collins supplies the avionics and acts as the single source for the complete SATCOM system. At the present time, AIRLINK® has been commissioned on large numbers of commercial and general aviation aircraft, including all versions of the Boeing 7XX series aircraft and Gulfstream II, III, and IV. The backlog of additional installations is significant, following the cycles of new aircraft deliveries and airline retrofit planning.

The antenna system represents mature microstrip phased array technology at Ball Aerospace. The key design issues involved execution of a low-cost producible package, while providing the required performance, reliability and maintainability. Some of these issues are discussed in the following sections. The antenna concepts are not difficult, but the low-cost producible design has allowed Ball to become the leader in this market.

2.0 AIRLINK® SYSTEM DESCRIPTION

AIRLINK® is an airborne satellite communications system fully compliant with multi-organizational standards such as ARINC 741, FAA, RTCA-DO-160C and INMARSAT multi-channel voice/data technical requirements. The system capability enables an aircraft to use a global aeronautical communication

network that consists of three components, the airborne earth station (AIRLINK®), geostationary satellites, and ground earth stations. The space segment consists of four INMARSAT satellites distributed in geosynchronous orbits that cover the major ocean regions as shown in Fig. 1. This distribution of satellite positions provides for worldwide SATCOM coverage except in some areas of the polar regions. The ground segment consists of three COMSAT Ground Earth Stations that interconnect with the Public Switched Telephone Network, providing worldwide service area coverage for the INMARSAT satellites. Aircraft/satellite communications are at L-band, while the ground station/satellite communications are at C-band.

Fig. 2. is a diagram of the AIRLINK® antenna system, consisting of two sidemounted apertures conformal to the aircraft skin, and six internally-mounted RF modules with associated cabling and connections for RF, power and control. The passive radiating apertures are connected by RF cable bundles to the Beam Steering Units (BSU), which contain 3-bit phase shifters for each radiating element, as well as an amplitude-tapered power divider/combiner for each aperture. The common port of the BSU is connected by a short cable run to the LNA/Diplexer module where the transmit and receive paths are split. Downstream from the LNA a simple starboard/port combiner leads to the main receive port. The transmit signal from the High Power Amplifier (HPA) is switched between starboard and port signal paths by means of a high power RF switch.

The architecture shown in this diagram is driven by the requirement for minimum drag from the radiating aperture, along with the availability of space inside the aircraft for mounting the RF electronics modules separate from the aperture. The resulting configuration provides a high degree of maintainability and reliability at the lowest possible

cost. Performance is compromised somewhat because of the passive BSU circuitry and cabling between the aperture and LNA/diplexer.

The coverage plan for the two sidemounted apertures is shown in Fig. 3. At a mounting angle of 45°, the aperture coverage overlaps at the zenith in a region (termed hysteresis) where a handoff routine between apertures is implemented. As expected, low elevation coverage on the nose and tail is limited.

3. DUAL USE

In a more recent internal development at Ball Aerospace, the AIRLINK® system has been reconfigured for secure military communications. The system is designed to offer worldwide, clear, reliable, two-way, air-to-ground or ground-to-air secure or non-secure voice and/or data communication. The secure interface unit of the audio/data subsystem performs analog/digital signal conversions between the Secure Terminal Unit (STU-III) and the Satellite Data Units of the SATCOM transceiver. The STU-III is an analog voice signal device used for both clear and secure voice and data communications.

The system is currently installed on a USAF Avionics Test Bed aircraft for operations testing and evaluation. Numerous deployments of the system are pending. Furthermore, the system is available for demonstration on a Ball Aerospace mobile test bed, which can demonstrate secure voice, video, FAX and PC transmissions.

4. APERTURE DESCRIPTION

The multilayer printed circuit antenna aperture is shown in Fig. 4, in which the circuitry on four different layers are superimposed. It can be readily seen that the full available

aperture is not used, resulting in a some compromise in performance. The space is used instead for the RF signal lines from the bundled coaxial inputs to each radiating element. As shown below, this choice enhances the low cost and producibility of the multilayer aperture. Both RF signal lines and polarization hybrids are implemented in buried microstrip, and are unshielded in the radiating aperture. The radiating elements themselves are two-layer stacked microstrip patches, covering the full range of transmit and receive bandwidth (about 10%).

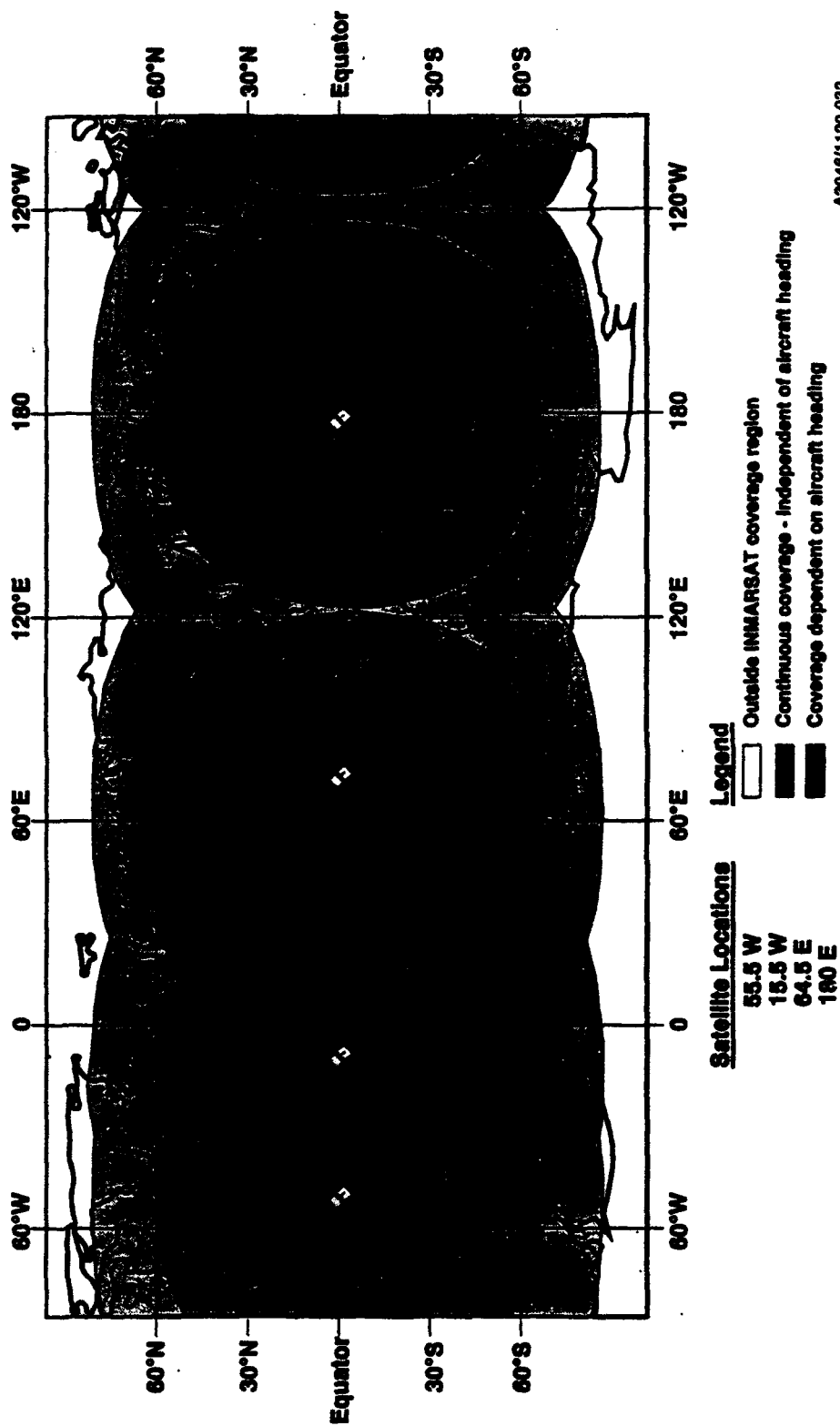
Fig. 5. is a cross-sectional view of the radiating aperture, having a total thickness 0.375 in. The bottom layer contains the RF signal lines and the polarization hybrids, fed in a surface launch from the coaxial connectors in two bundles. Within the polarization hybrid the signal actually jumps from the trace layer on the bottom board to the back side of the lower (driver) element board, so that the feed ribbon can be installed easily prior to any laminations. The upper (driven) element is etched on a separate board, and the entire four-layer lamination (with a thin laminated radome) is accomplished at one time.

The aperture assembly is about as simple as a multilayer microstrip antenna can be. The installation of the two coaxial connector bundles (shown in Fig. 6) is actually the most difficult part of the process. There were certainly design alternatives that would have provided marginally greater performance, but at higher levels of complexity and production cost.

5. PERFORMANCE DATA

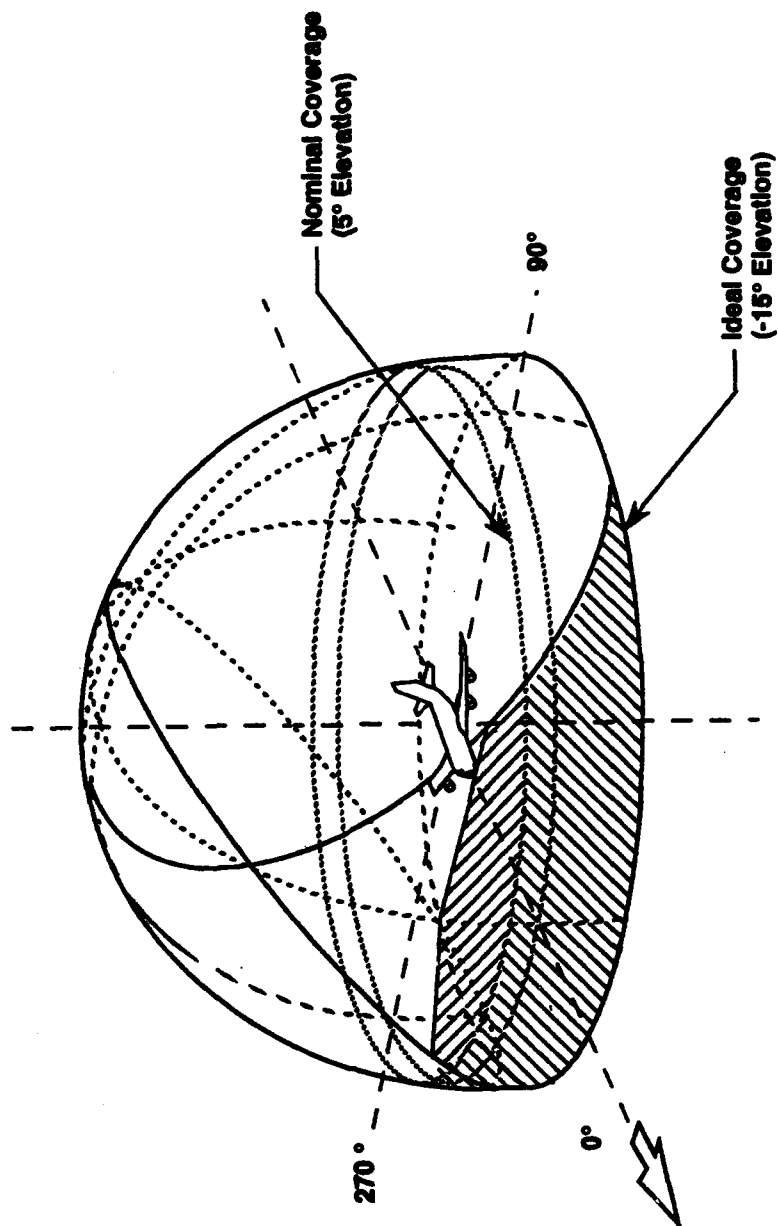
Pattern, gain, G/T and ERP data for the AIRLINK® antenna subsystem are shown in the following figures. Fig. 7. is a typical pattern plot with the antenna scanned to 45° along the

long (azimuth) dimension of the array. Testing and qualification of AIRLINK® also requires patterns where the antenna is constantly scanning as it rotates, so that the peak of beam points to the same angle. An example of this type of peak gain pattern is shown in Fig. 8. The same type of scanning also produces contour plots from data taken over half-space. Fig. 9 is a plot of the 12.0 dBic gain contour for AIRLINK® S/N 120. Single antenna data is also transformed into aircraft coordinates for particular commissionings to determine ERP and G/T coverage. Fig. 10 and 11 show aircraft coverage contours at 25.5 dBW ERP and -13.0 dB/K° G/T.



A3046/1120.032

Fig. 1. AIRLINK® GLOBAL COVERAGE



A3048/1120.06/1

Fig. 3. AIRLINK® COVERAGE DIAGRAM

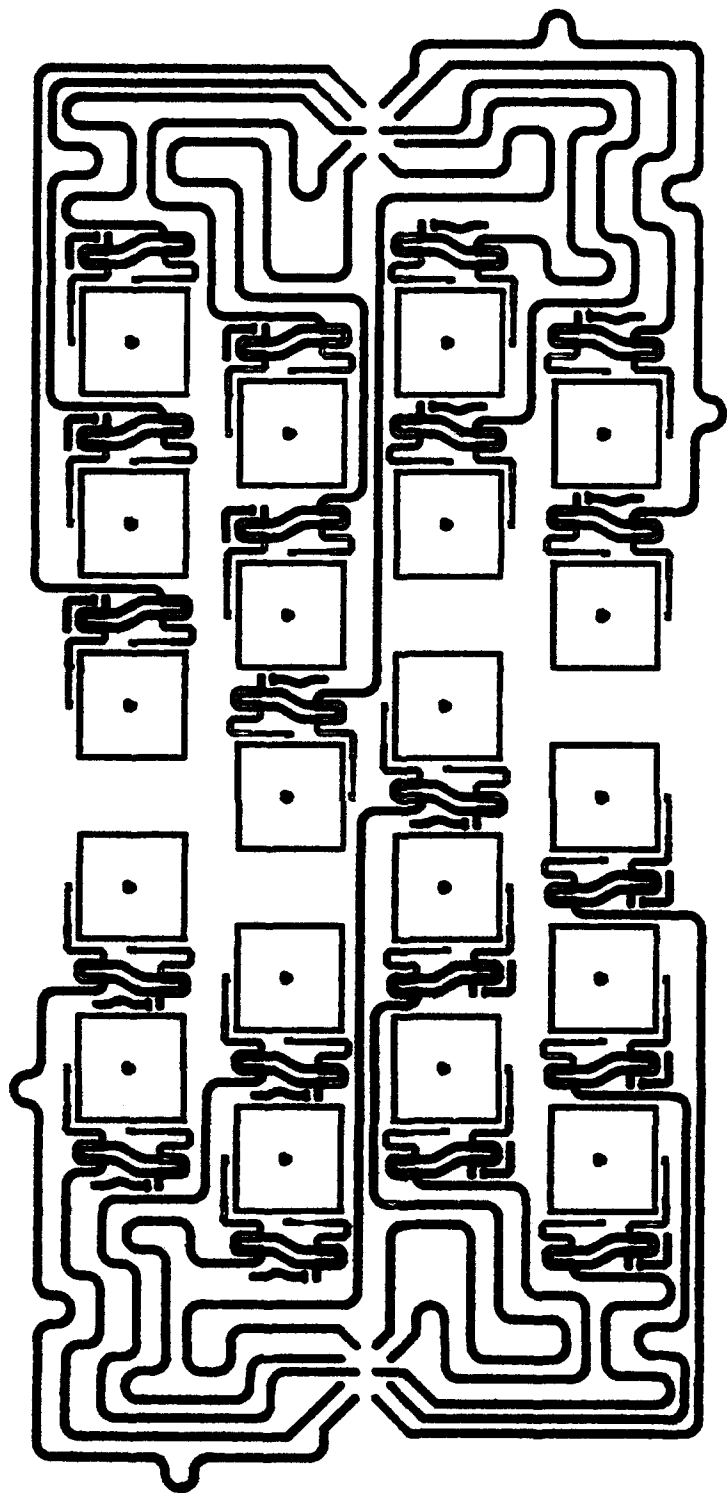


Fig. 4. AIRLINK● APERTURE LAYOUT

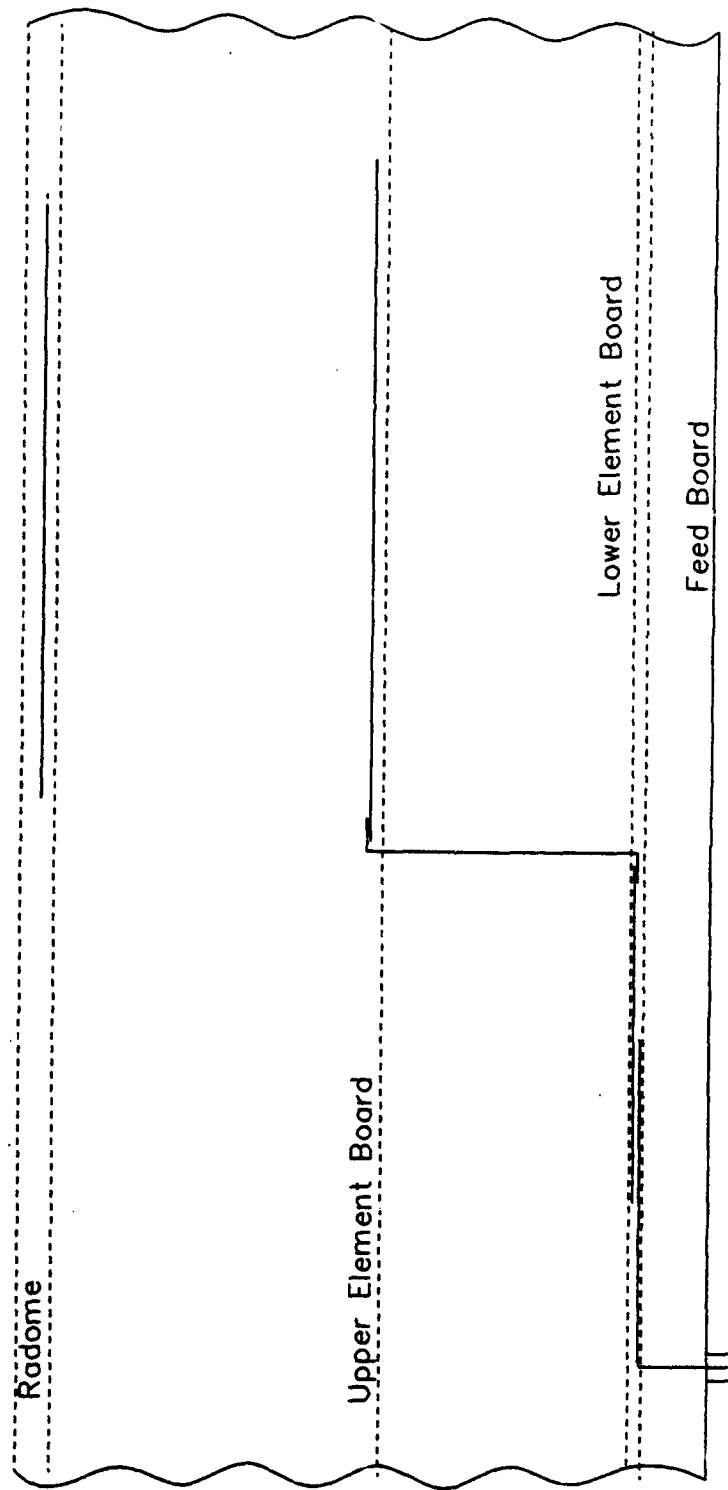
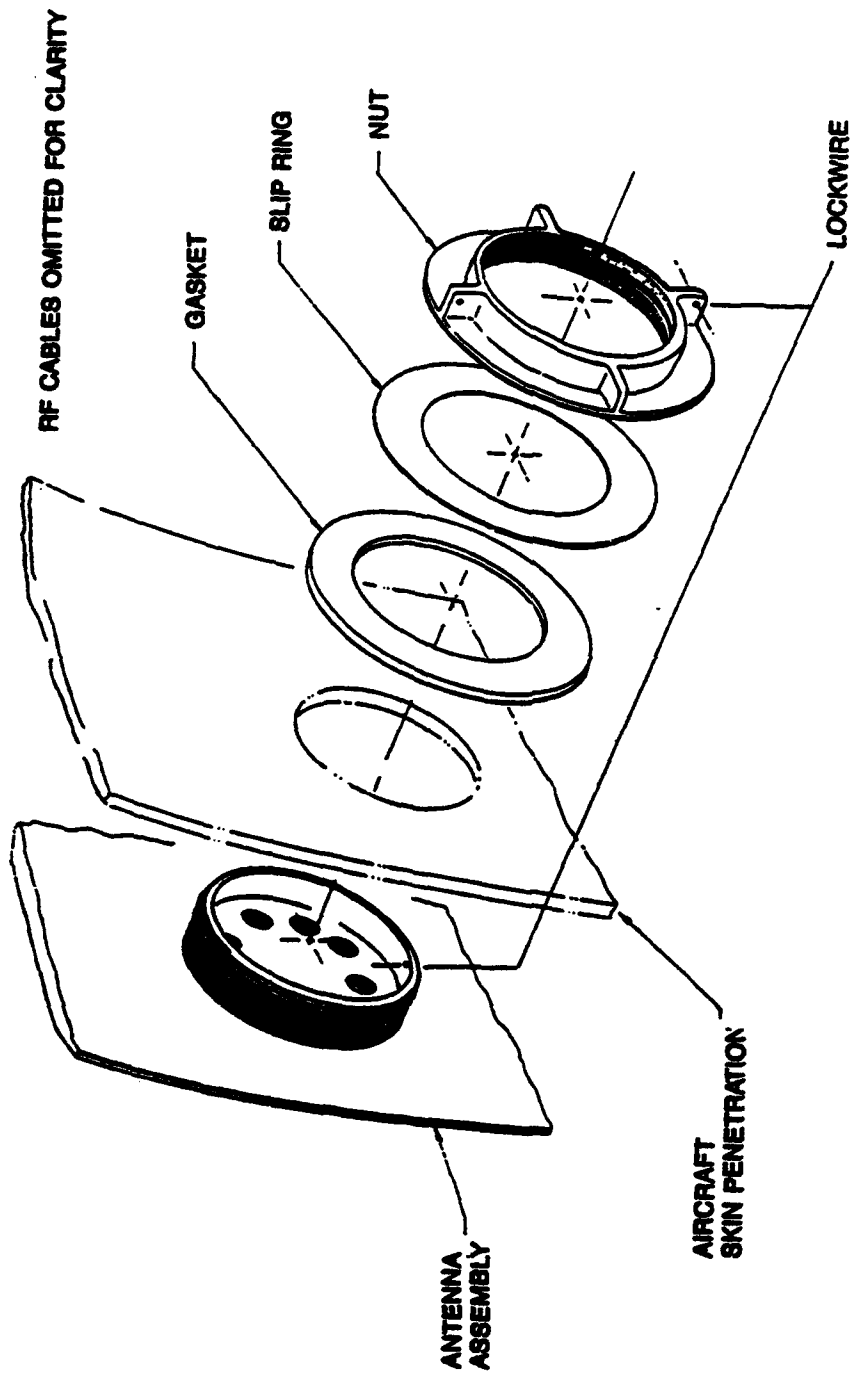


Fig. 5. APERTURE CROSS SECTIONAL VIEW



AN 6589-2A3

Fig. 6. RF CABLE CONNECTOR ASSEMBLY

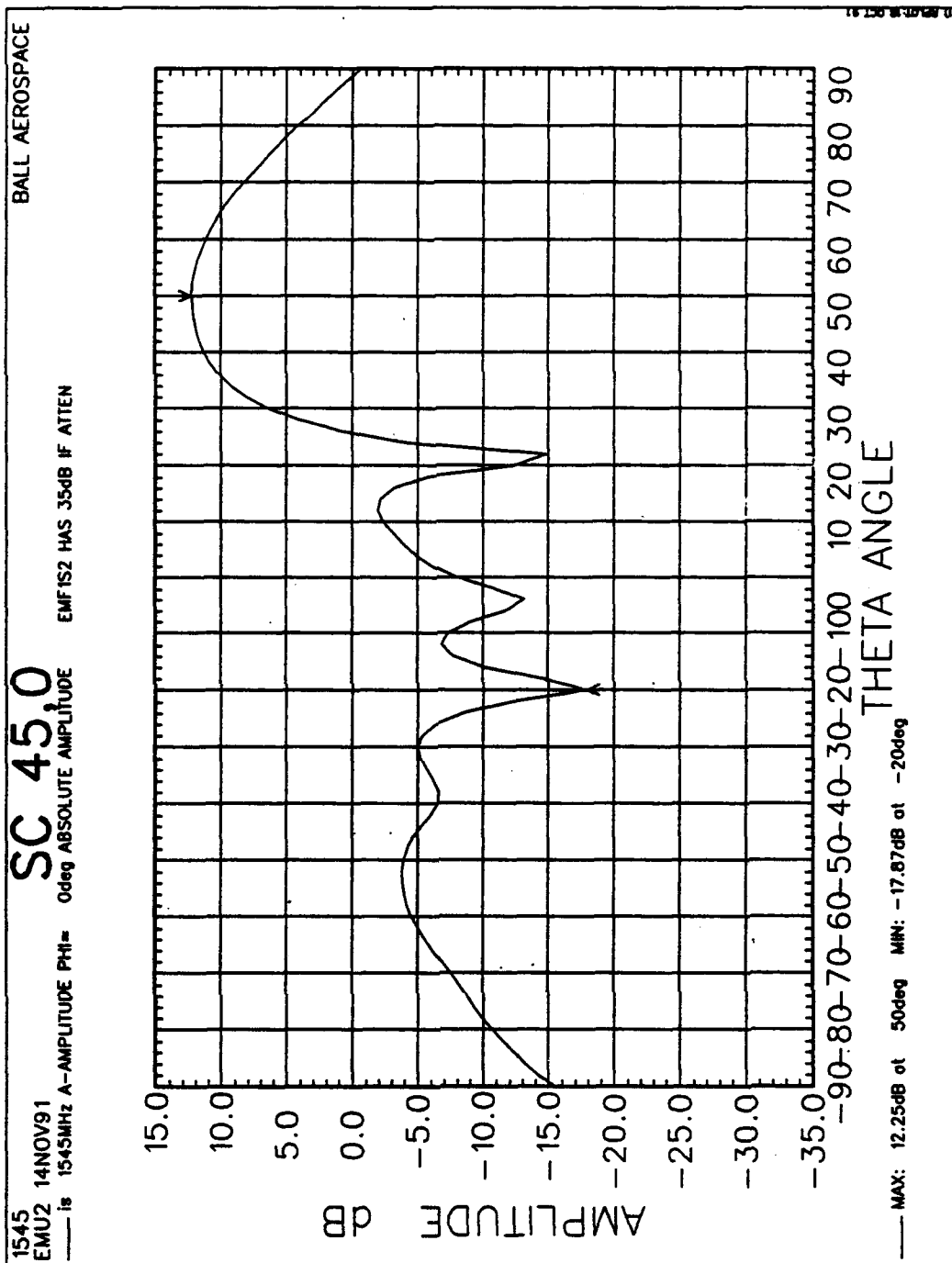


Fig. 7. AIRLINK® PATTERN SCANNED 45°

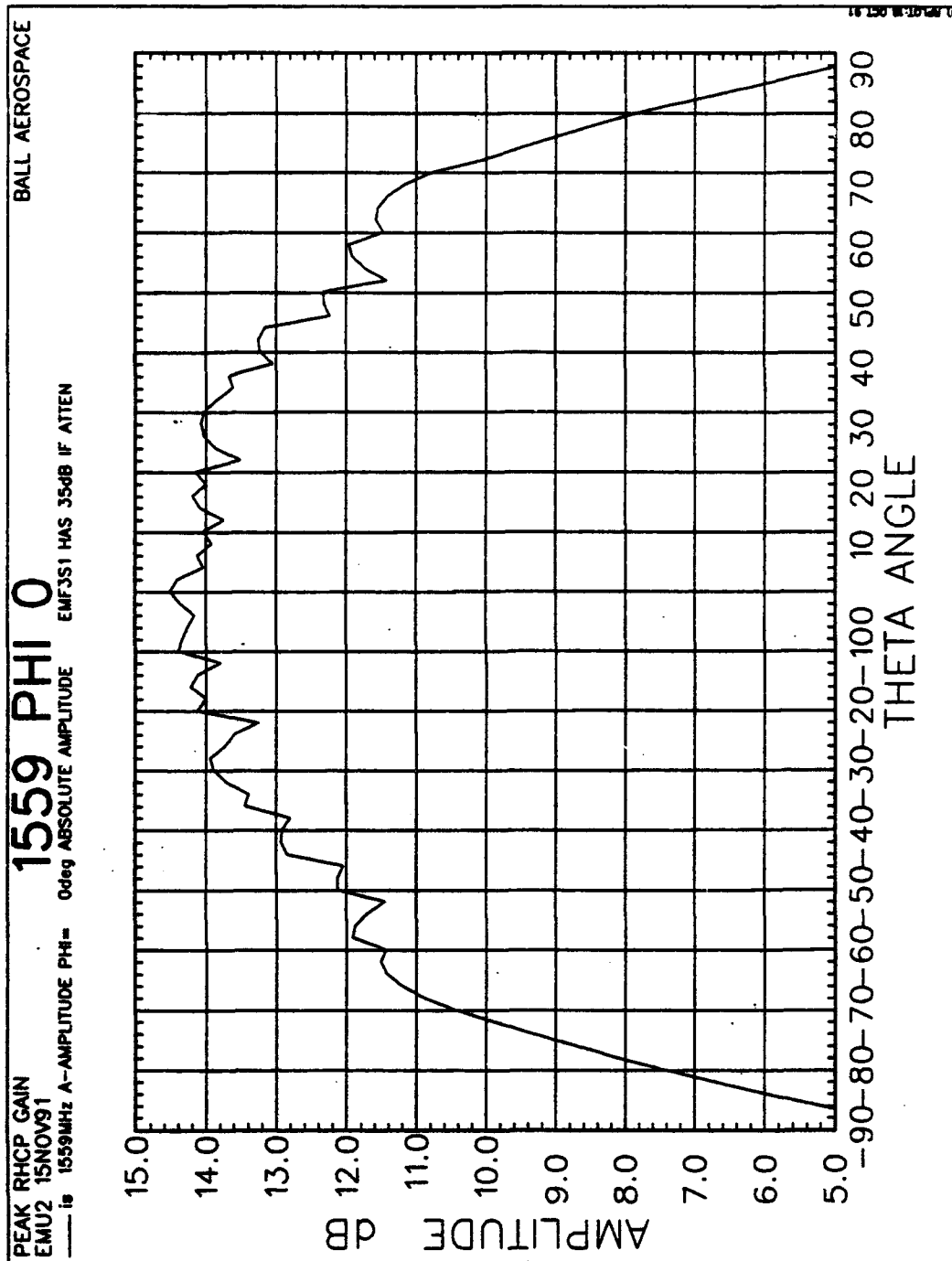


Fig. 8. PEAK GAIN PATTERN, AZ=0°

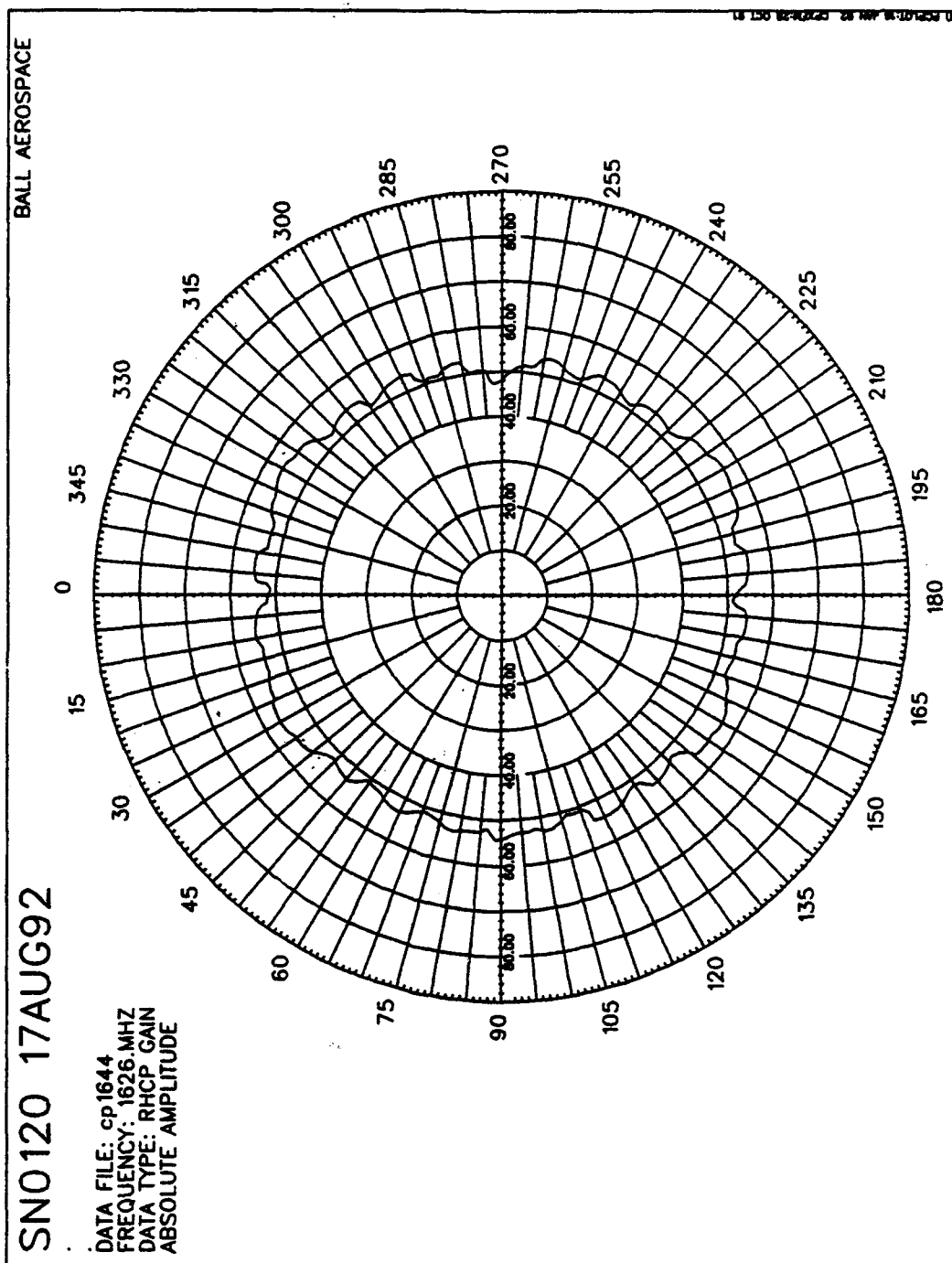


Fig. 9. AIRLINK® 12.0 dBic CONTOUR

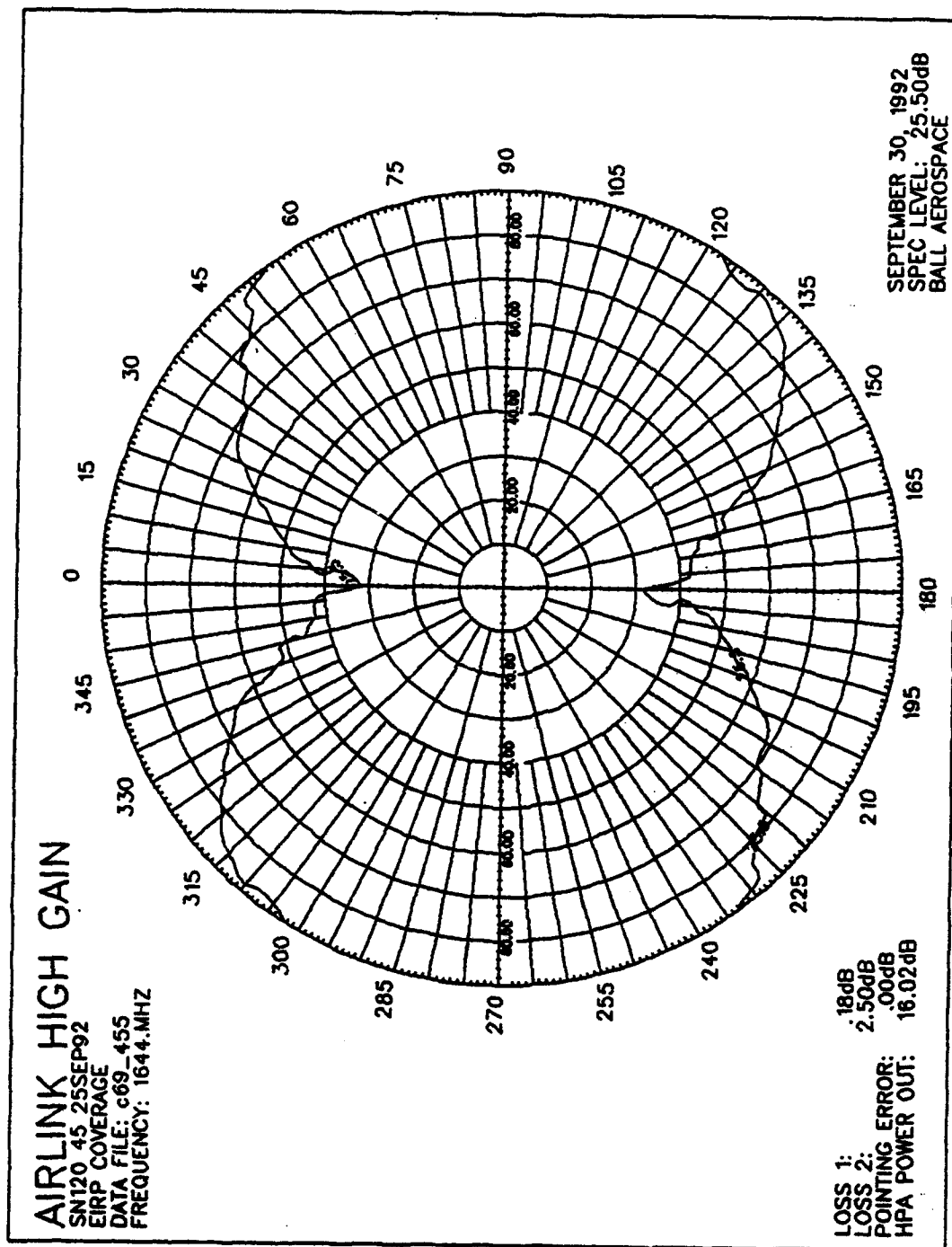


Fig. 10. AIRLINK® 25.5 dBW CONTOUR IN AIRCRAFT COORDINATES

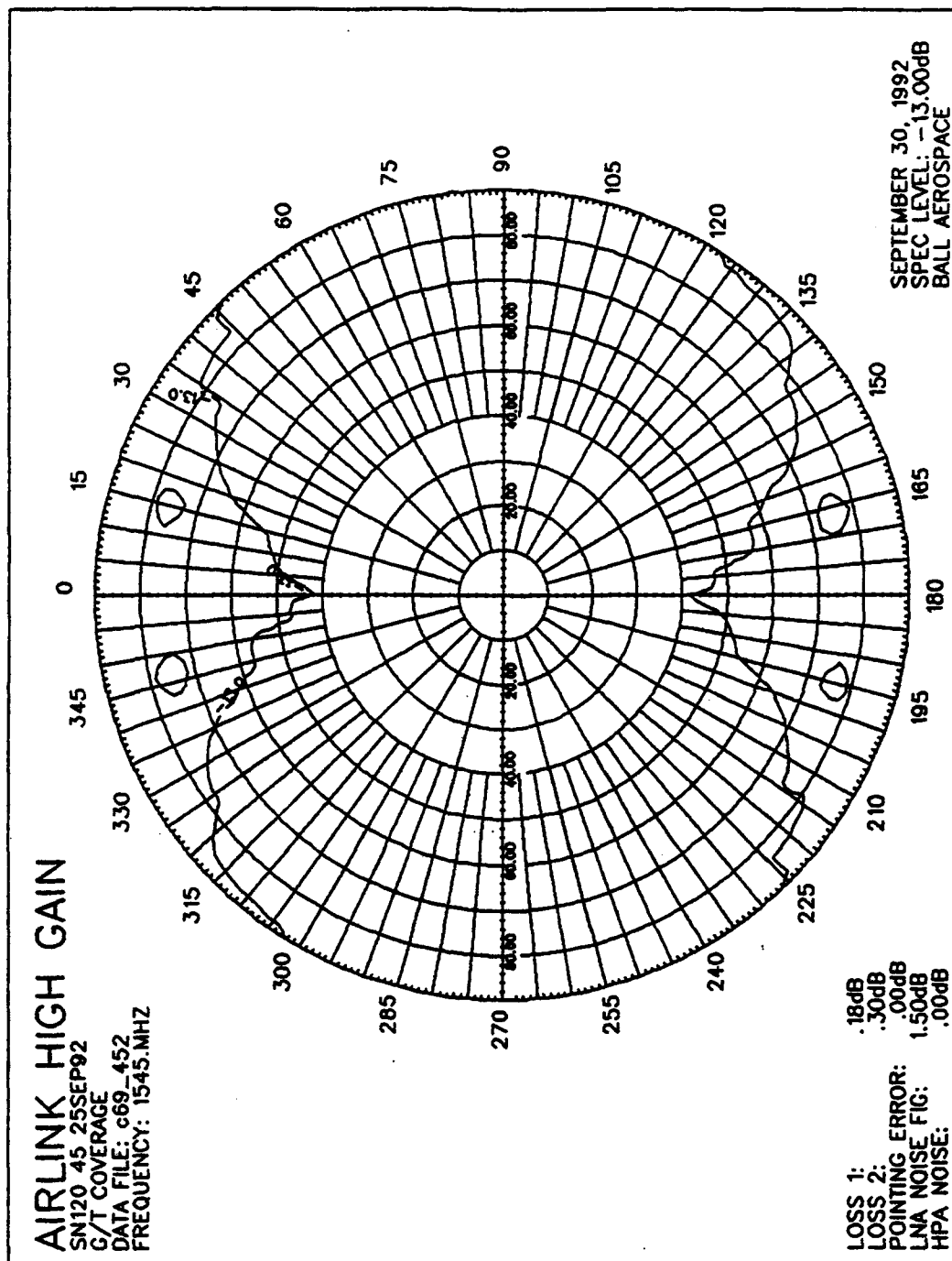


Fig. 11. AIRLINK -13.0 dB/K° CONTOUR IN AIRCRAFT COORDINATES